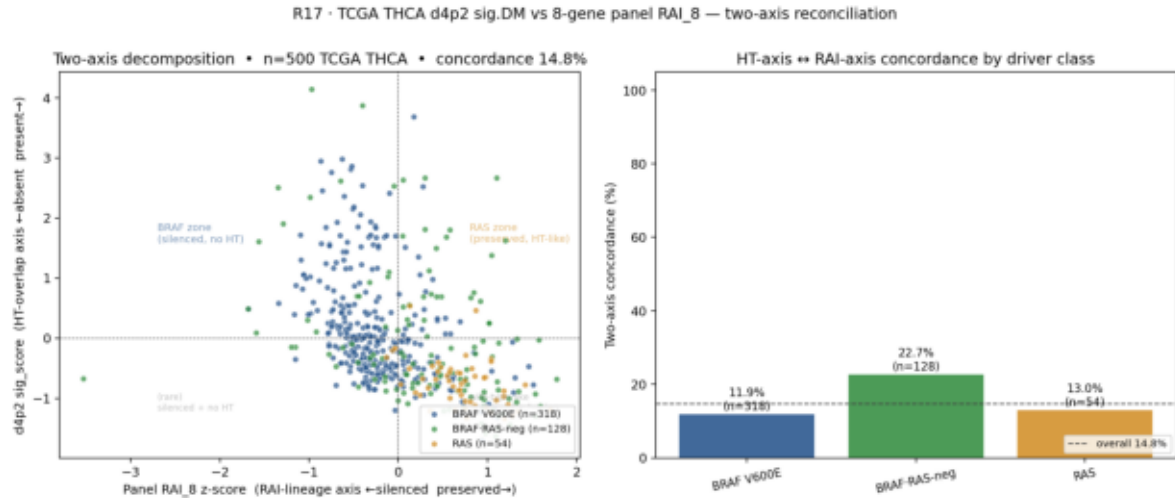


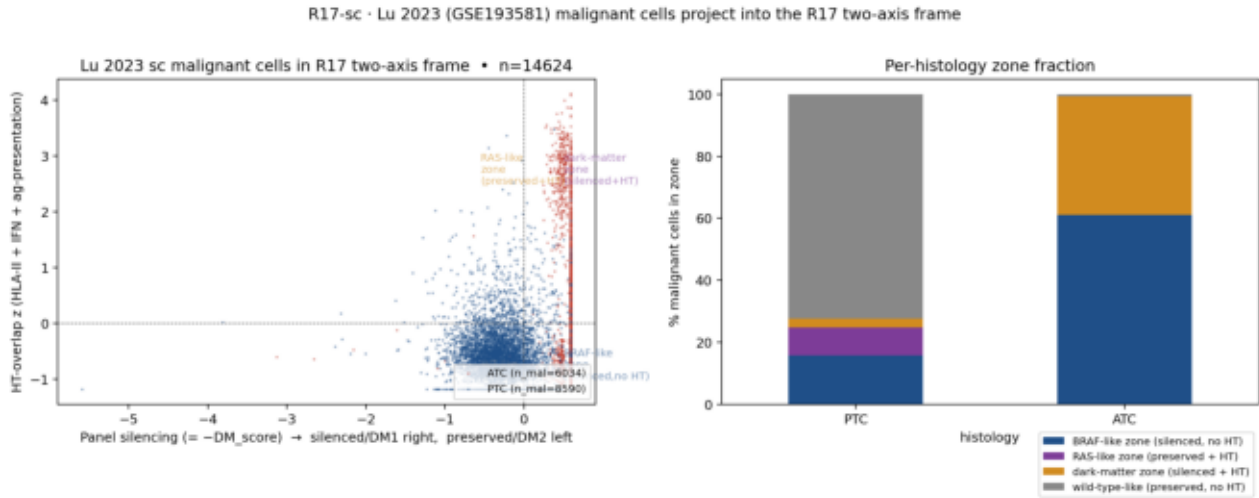
Discovery context — an 8-gene differentiation-silencing axis across 4 data pillars
RNA bulk · single-cell · proteome · methylation (Paper 1 R17 4-pillar reuse)

a · TCGA bulk RNA (n = 500)



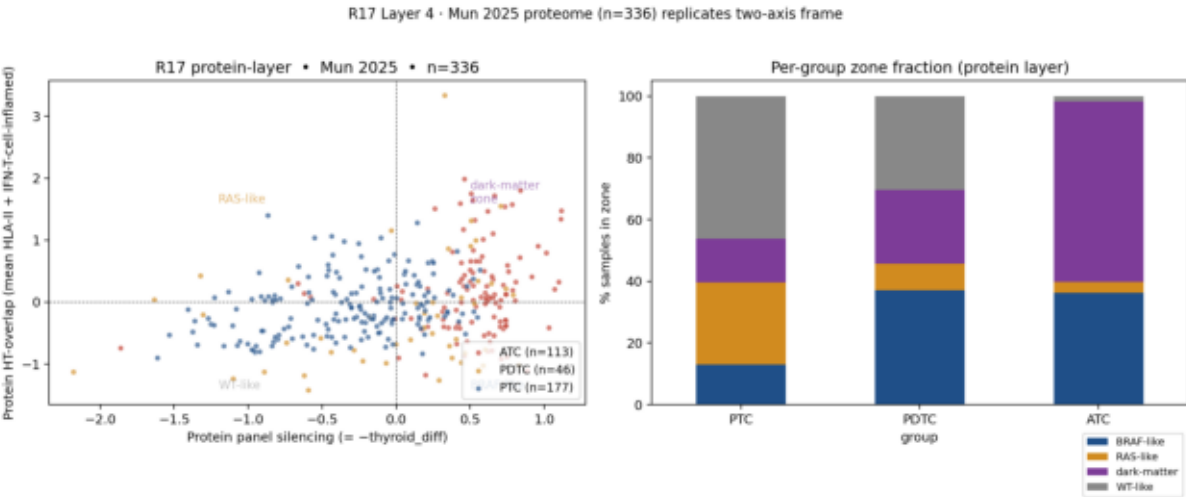
Two-axis decomposition · Tier 4 proxy · 14.8 % d4p2↔panel label concordance

b · Lu 2023 single-cell (n = 14,624)



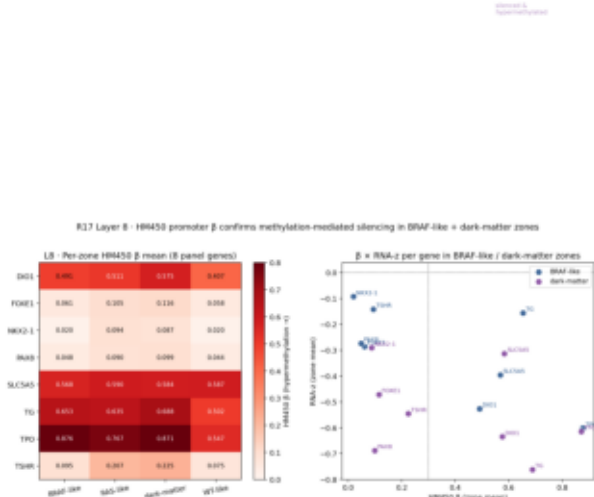
ATC 38.3 % silenced + HT-overlap cells · cellular substrate

c · Mun 2025 proteome (n = 336)



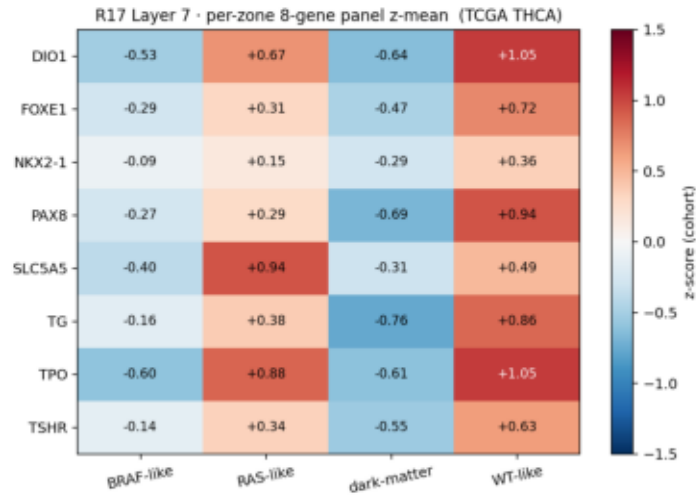
ATC 58.4 % silenced + HT zone · Fisher OR = 8.54 $p = 2.7 \times 10^{-15}$

d · TCGA HM450 β (n = 518)



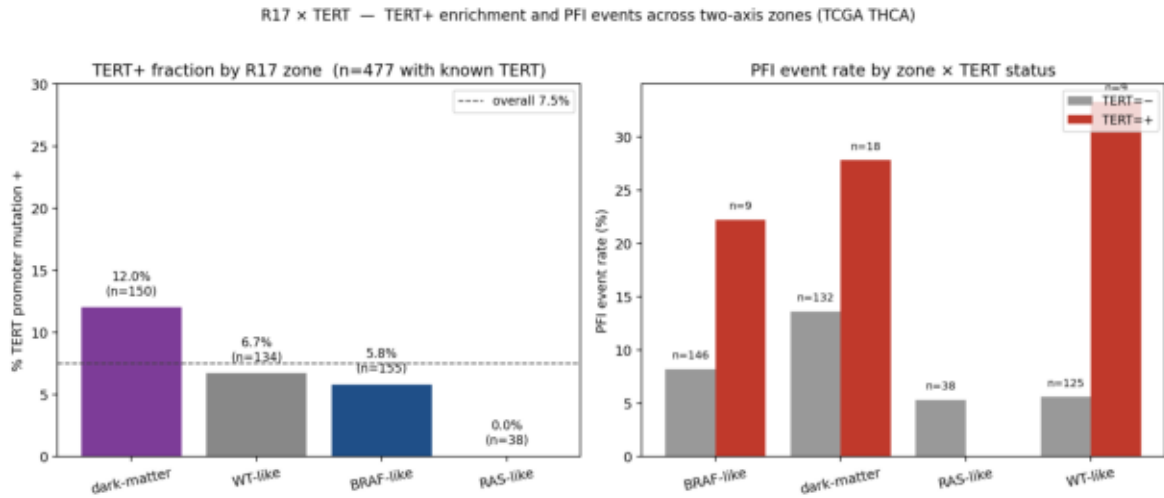
Methylation-mediated silencing in 4/8 panel genes · DIO1·SLC5A5·TG·TPO

e · Per-zone 8-gene profile



Two silencing programs: BRAF-like → TPO/DIO1, dark-matter → TG/PAX8/TSHR

f · TERT × zone (n = 477)



TERT+ uniquely enriched in dark-matter zone · OR = 2.34 $p = 0.016$